VIVEKANANDHA SAINIK COACHING CENTER

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ALL INDIA SAINIK SCHOOLS ENTRANCE EXAMINATION (AISSEE) — CLASS VI

MOCK QUESTION PAPER — 01

DURATION: 150 MINUTES (2½ HOURS) | MAXIMUM MARKS: 300

Candidate Name: _____ **Roll Number:** _____ **OMR Sheet No.:** _____

PART A — EXAMINATION SCHEME & CANDIDATE INSTRUCTIONS

1. This booklet contains 125 Multiple Choice Questions divided into four sections: **Section A: Language (English)** (Q.1–25, 2 Marks each = 50 Marks); **Section B: Mathematics** (Q.26–75, 3 Marks each = 150 Marks); **Section C: Intelligence** (Q.76–100, 2 Marks each = 50 Marks); **Section D: General Knowledge** (Q.101–125, 2 Marks each = 50 Marks).
2. All questions are compulsory. There is **no negative marking** for wrong answers.
3. Darken only one circle on your OMR Answer Sheet using a Black or Blue ballpoint pen.

SECTION A : ENGLISH LANGUAGE (Q.1 TO 25)

Q.1. Choose the most appropriate collective noun to complete the sentence: A _____ of bees attacked the intruder.

(A) flock (B) swarm
(C) herd (D) pack

Q.2. Fill in the blank with the correct preposition: She has been living in Chennai _____ 2018.

(A) for (B) since
(C) from (D) during

Q.3. Choose the correct article to fill in the blank: Copper is _____ useful metal.

(A) a (B) an
(C) the (D) no article

Q.4. Select the correct relative pronoun: The girl _____ won the gold medal is my cousin.

(A) which (B) who
(C) whom (D) whose

Q.5. Fill in the blank with the correct degree of adjective: Platinum is _____ than gold.

(A) costly (B) more costly
(C) costliest (D) most costly

Q.6. Identify the correct tense form: Look! The children _____ in the garden.

(A) play (B) are playing
(C) played (D) have played

Q.7. Choose the correctly spelt word from the given options:

(A) Accomodation (B) Accommodation
(C) Acommodation (D) Accomodatoin

Q.8. Choose the word nearest in meaning (Synonym) to COURTEOUS:

(A) Rude (B) Polite
(C) Proud (D) Clever

Q.9. Choose the word opposite in meaning (Antonym) to ANCIENT:

(A) Old (B) Antique
(C) Modern (D) Historic

Q.10. Choose the correct meaning of the idiom 'Burn the midnight oil':

(A) To waste fuel (B) To study or work late into the night
(C) To set fire at night (D) To sleep early

Q.11. Complete the sentence with a suitable question tag: You aren't attending the class tomorrow, _____?

(A) aren't you (B) do you
(C) are you (D) won't you

Q.12. What is the feminine gender form of 'Bachelor'?

(A) Spinster (B) Widow

(C) Governess (D) Dame

Q.13. Select the correct plural form of 'Hypothesis':

(A) Hypothesises (B) Hypotheses

(C) Hypothesiss (D) Hypothesi

Q.14. Identify the verb in the sentence 'Rohan kicked the football' and its type:

(A) Rohan (Noun) (B) kicked (Transitive verb)

(C) football (Object) (D) the (Article)

Q.15. In the sentence 'The soldier fought bravely', identify the underlined adverb type ('bravely'):

(A) Adverb of Time (B) Adverb of Place

(C) Adverb of Manner (D) Adverb of Degree

Q.16. Choose the correct conjunction: Work hard _____ you will fail in the examination.

(A) and (B) or

(C) so (D) because

Q.17. Rearrange the jumbled words to form a meaningful sentence: P: always / Q: speaks / R: the truth / S: He

(A) PQRS (B) SPQR

(C) PSRQ (D) RSPQ

Q.18. Choose the one word substitute for: 'A person who studies stars, planets, and celestial bodies':

(A) Astronaut (B) Astronomer

(C) Astrologer (D) Aviator

Q.19. Choose the appropriate interjection to express sudden grief or sorrow:

(A) Hurrah! (B) Bravo!

(C) Alas! (D) Eureka!

Q.20. Choose the correct confusing word: The heavy storm had a severe _____ on the standing crops.

(A) affect (B) effect

(C) affection (D) effective

READING COMPREHENSION PASSAGE

Read the following passage carefully and answer questions 21 to 25: The Great Banyan Tree in the Acharya Jagadish Chandra Bose Indian Botanic Garden, Howrah, near Kolkata, is over 250 years old. Visually, it resembles a dense forest rather than a single tree. It has over 3,600 prop roots that drop down from branches and root themselves into the ground, acting like supporting pillars. The main trunk of the tree was infected by a deadly fungus after two severe cyclones in 1864 and 1867, and had to be surgically removed in 1925 to save the rest of the tree. Despite having no central trunk, the tree continues to thrive and covers an area of nearly 3.5 acres.

Q.21. According to the passage, where is the Great Banyan Tree located?

(A) New Delhi (B) Howrah, near Kolkata

(C) Chennai (D) Bengaluru

Q.22. What function do the 3,600 prop roots serve for the tree?

(A) They absorb sunlight (B) They act like supporting pillars

(C) They produce sweet flowers (D) They protect against insects

Q.23. Why was the main central trunk of the tree surgically removed in 1925?

(A) To make timber furniture (B) It was infected by a deadly fungus

(C) To clear space for roads (D) It fell down naturally

Q.24. What caused the initial damage that led to the fungal infection of the tree?

(A) Severe droughts (B) Forest fires

(C) Two severe cyclones in 1864 & 1867 (D) Heavy frost

Q.25. Which word in the passage means 'to grow vigorously or healthily'?

(A) Resemble (B) Infect

(C) Thrive (D) Surgical

SECTION B : MATHEMATICS (Q.26 TO 50)

Q.26. Find the difference between the greatest and smallest 5-digit numbers formed using the digits 7, 1, 9, 0, and 4 (each used only once):

(A) 86,931 (B) 87,031

(C) 86,831 (D) 87,931

Q.27. Find the sum of the place value and the face value of the digit 8 in the number 5,82,439:

(A) 80,008 (B) 8,008

(C) 800,008 (D) 80,080

Q.28. Convert the Roman numeral CLXXVIII into Hindu-Arabic numeral system:

(A) 168 (B) 178

(C) 188 (D) 158

Q.29. Find the Highest Common Factor (HCF) of 84, 126, and 210:

(A) 21 (B) 42

(C) 14 (D) 28

Q.30. Four bells toll together at intervals of 6, 8, 12, and 18 seconds respectively. After how many seconds will they toll together next?

(A) 36 seconds (B) 72 seconds

(C) 144 seconds (D) 54 seconds

Q.31. Simplify using BODMAS rule: $75 - [20 + \{30 - (12 \div 4 \times 2)\}]$

(A) 31 (B) 35

(C) 27 (D) 41

Q.32. Evaluate: $(3/5 + 1/2) \div (7/10 - 1/4)$

(A) 2 4/9 (B) 1 5/9

(C) 2 2/9 (D) 3 1/9

Q.33. Arrange the fractions 4/5, 7/10, 2/3, and 5/6 in descending order:

(A) 5/6, 4/5, 7/10, 2/3 (B) 4/5, 5/6, 7/10, 2/3

(C) 5/6, 7/10, 4/5, 2/3 (D) 2/3, 7/10, 4/5, 5/6

Q.34. Convert the decimal number 0.0325 into a fraction in its lowest terms:

(A) 13/400 (B) 7/200

(C) 31/800 (D) 13/200

Q.35. If 15 workers can complete a wall construction in 24 days, how many days will 20 workers take to finish the same work?

(A) 18 days (B) 20 days

(C) 16 days (D) 32 days

Q.36. Divide ■ 3,600 among A, B, and C in the ratio 3 : 4 : 5. What is B's share?

(A) ■ 900 (B) ■ 1,200

(C) ■ 1,500 (D) ■ 1,000

Q.37. An article was purchased for ■ 450 and sold for ■ 540. Calculate the gain percentage:

(A) 15% (B) 18%

(C) 20% (D) 25%

Q.38. By selling a bicycle for ■ 2,850, a shopkeeper incurs a loss of 5%. What was the cost price of the bicycle?

(A) ■ 2,950 (B) ■ 3,000

(C) ■ 3,100 (D) ■ 3,200

Q.39. The average weight of 6 boys is 42 kg. When a new boy joins the group, the average weight becomes 44 kg. Find the weight of the new boy:

(A) 50 kg (B) 52 kg

(C) 54 kg (D) 56 kg

Q.40. In a school of 800 students, 35% are girls. How many boys are enrolled in the school?

(A) 280 (B) 520

(C) 480 (D) 540

Q.41. Find the simple interest on ■ 8,000 at 7.5% per annum for 3 years:

(A) ■ 1,600 (B) ■ 1,800

(C) ■ 2,100 (D) ■ 1,750

Q.42. In how many years will a sum of ■ 5,000 double itself at 10% per annum simple interest?

(A) 8 years (B) 10 years

(C) 12 years (D) 5 years

Q.43. An angle is 24° greater than its complementary angle. Find the measure of the angle:

(A) 57° (B) 33°

(C) 66° (D) 48°

Q.44. Two supplementary angles are in the ratio 4 : 5. Find the measure of the smaller angle:

(A) 80° (B) 100°

(C) 70° (D) 90°

Q.45. In a triangle ABC, ∠A = 55° and ∠B = 65°. Find the measure of ∠C:

(A) 50° (B) 60°

(C) 70° (D) 65°

Q.46. Find the perimeter of a rectangle whose area is 300 cm² and length is 25 cm:

(A) 68 cm (B) 74 cm

(C) 82 cm (D) 70 cm

Q.47. A square garden of side 24 m is surrounded by a path 2 m wide outside it. Find the area of the path:

(A) 196 m² (B) 208 m²

(C) 216 m² (D) 184 m²

Q.48. Find the cost of fencing a circular park of radius 14 m at the rate of ■ 15 per metre (Use $\pi = 22/7$):

(A) ■ 1,240 (B) ■ 1,320

(C) ■ 1,480 (D) ■ 1,180

Q.49. A wooden cuboid measures 12 cm × 10 cm × 8 cm. Find its Total Surface Area (TSA):

(A) 480 cm² (B) 592 cm²

(C) 612 cm² (D) 576 cm²

Q.50. How many cubical blocks of side 4 cm can be cut from a wooden block of dimensions 32 cm × 24 cm × 16 cm?

(A) 160 (B) 192

(C) 216 (D) 128

Q.51. Convert the speed of 72 km/hr into metres per second (m/s):

(A) 15 m/s (B) 20 m/s

(C) 25 m/s (D) 18 m/s

Q.52. A train 150 m long passes a signal pole in 9 seconds. Find the speed of the train in km/hr:

(A) 54 km/hr (B) 60 km/hr

(C) 65 km/hr (D) 72 km/hr

Q.53. Solve for x: $5x - 7 = 3x + 11$

(A) 7 (B) 8

(C) 9 (D) 10

Q.54. If $x = 3$ and $y = -2$, evaluate the algebraic expression: $2x^2 - 3xy + y^2$

(A) 32 (B) 36

(C) 40 (D) 44

Q.55. The sum of three consecutive natural numbers is 87. Find the greatest number among them:

(A) 28 (B) 29

(C) 30 (D) 31

Q.56. Convert 45° Celsius into Fahrenheit scale (°F):

(A) 105°F (B) 113°F

(C) 121°F (D) 117°F

Q.57. Perform the integer operation: (−38) + (+45) + (−27) + (+60)

(A) +30 (B) +40

(C) −40 (D) +50

Q.58. Subtract (–24) from (–15):

(A) –39 (B) +9

(C) –9 (D) +39

Q.59. How many prime numbers lie between 40 and 70?

(A) 6 (B) 7

(C) 8 (D) 9

Q.60. Which of the following numbers is divisible by 11?

(A) 38,412 (B) 47,215

(C) 52,914 (D) 61,835

Q.61. Find the perimeter of a regular octagon of side 6.5 cm:

(A) 45.5 cm (B) 52 cm

(C) 58.5 cm (D) 48 cm

Q.62. A milk container holds 6.5 litres of milk. How many glasses of 250 mL capacity each can be filled completely?

(A) 24 (B) 26

(C) 28 (D) 30

Q.63. Round off 48,762 to the nearest thousand:

(A) 48,000 (B) 49,000

(C) 48,800 (D) 50,000

Q.64. An angle measuring 215° is classified as a/an:

(A) Acute angle (B) Obtuse angle

(C) Reflex angle (D) Straight angle

Q.65. How many lines of symmetry does a regular pentagon have?

(A) 4 (B) 5

(C) 10 (D) Infinite

Q.66. Find the mean (average) of the first six prime numbers:

(A) 6.83 (B) 7.16

(C) 6.50 (D) 7.50

Q.67. What percentage of 4 kg is 250 grams?

(A) 5.25% (B) 6.25%

(C) 7.50% (D) 8.00%

Q.68. In a circle, a chord that passes directly through the center is called its:

(A) Radius (B) Arc

(C) Diameter (D) Sector

Q.69. Read the two statements regarding geometric figures: Statement I: Every square is also a rectangle. Statement II: Diagonals of a rhombus are always equal in length.

(A) Both I and II are true (B) Both I and II are false

(C) Statement I is true, II is false (D) Statement I is false, II is true

Q.70. The ratio of two numbers is 5 : 8 and their HCF is 9. Find the larger number:

(A) 45 (B) 64

(C) 72 (D) 81

Q.71. A fruit seller buys 40 kg apples for ■ 2,400 and sells them at ■ 72 per kg. Find his net gain or loss:

(A) ■ 480 Gain (B) ■ 480 Loss

(C) ■ 360 Gain (D) ■ 520 Gain

Q.72. Find the value of: 3/8 + 5/12 – 1/6

(A) 1/2 (B) 5/8

(C) 3/4 (D) 7/12

Q.73. The length and breadth of a rectangular ground are in ratio 7 : 4. If its perimeter is 220 m, find its length:

(A) 40 m (B) 70 m

(C) 80 m (D) 60 m

Q.74. A car covers 180 km in 4 hours. How far will it travel in 7 hours at the same speed?

(A) 280 km (B) 315 km

(C) 340 km (D) 295 km

Q.75. In a school library pictograph, 1 book symbol represents 15 books. How many symbols represent 105 books?

(A) 5 (B) 6

(C) 7 (D) 8

SECTION C : INTELLIGENCE (Q.76 TO 100)

Q.76. Find the missing term in the number series: 4, 9, 19, 39, 79, ?

(A) 149 (B) 158

(C) 159 (D) 169

Q.77. Complete the letter series: C, F, I, L, O, ?

(A) P (B) Q

(C) R (D) S

Q.78. Complete the verbal analogy: Newspaper : Press :: Cloth : ?

(A) Tailor (B) Mill

(C) Fibre (D) Cotton

Q.79. Complete the number analogy: 12 : 144 :: 15 : ?

(A) 210 (B) 225

(C) 240 (D) 256

Q.80. Choose the odd one out from the given alternatives:

(A) Copper (B) Silver

(C) Gold (D) Brass

Q.81. Choose the odd number out from the given options:

(A) 121 (B) 169

(C) 225 (D) 216

Q.82. In a certain code language, 'CAMEL' is written as 'ECNGN'. How is 'HORSE' written in that code?

(A) JQTUG (B) IPSTF

(C) JQTUF (D) IQTVG

Q.83. Pointing to a man, Neha said, 'His mother is the only daughter of my mother.' How is Neha related to the man?

(A) Sister (B) Mother

(C) Aunt (D) Daughter

Q.84. A person walks 10 km East, turns right and walks 5 km, then turns right again and walks 10 km. How far is he from his starting point?

(A) 10 km (B) 5 km

(C) 15 km (D) 25 km

Q.85. In a class of 40 students, Priya is ranked 12th from the top. What is her rank from the bottom?

(A) 28th (B) 29th

(C) 30th (D) 27th

Q.86. Arrange the words in alphabetical dictionary order: 1. Mango 2. Master 3. Market 4. Margin 5. Magnet

(A) 5, 1, 4, 3, 2 (B) 5, 4, 1, 3, 2

(C) 1, 5, 4, 3, 2 (D) 5, 1, 3, 4, 2

Q.87. Arrange the units in logical sequence from smallest to largest: 1. Kilometre 2. Millimetre 3. Metre 4. Centimetre

(A) 2, 4, 3, 1 (B) 4, 2, 3, 1

(C) 2, 3, 4, 1 (D) 2, 4, 1, 3

Q.88. If '+' means '÷', '–' means '×', '×' means '+', and '÷' means '–', evaluate: 36 + 6 – 3 × 8 ÷ 4

(A) 20 (B) 22

(C) 24 (D) 18

Q.89. Which word CANNOT be formed using the letters of 'TRANSPARENT'?

(A) PART (B) TRAIN

(C) PARENT (D) STREAM

Q.90. Find the missing number in the matrix pattern: Row 1: 3, 5, 16 | Row 2: 4, 6, 25 | Row 3: 5, 7, ?

(A) 35 (B) 36

(C) 37 (D) 40

Q.91. Select the option showing the mirror image of 'SAINIK' in a vertical mirror placed on its right.

(C) K I N I A S (D) S A I N I K (flipped)

Q.92. Which Venn diagram best represents the relationship between: Vegetables, Potato, Cabbage?

(A) Two separate circles inside a large circle (B) Three intersecting circles

(C) Three concentric circles (D) Two intersecting circles inside a large circle

Q.93. How many total triangles are there in a square divided by both diagonals?

(A) 4 (B) 6

(C) 8 (D) 10

Q.94. If 1st January 2024 was a Monday, what day of the week was 31st January 2024?

(A) Tuesday (B) Wednesday

(C) Thursday (D) Friday

Q.95. Which letter is 5th to the right of the 12th letter from the left in the English alphabet?

(A) P (B) Q

(C) R (D) O

Q.96. Statement: All cats are mammals. All mammals have lungs. Conclusion:

(A) All cats have lungs (B) No cat has lungs

(C) Some cats do not have lungs (D) Mammals are cats

Q.97. Complete the analogy: Circle : Sphere :: Square : ?

(A) Rectangle (B) Cube

(C) Pyramid (D) Cylinder

Q.98. If A = 1, B = 2... and CODE = 27 (3+15+4+5), what is the code value for EXAM?

(A) 41 (B) 43

(C) 45 (D) 42

Q.99. Five friends A, B, C, D, E sit in a row facing North. C is in middle, A is right of C, E is at left end, B is between E and C. Who is at right end?

(A) A (B) B

(C) D (D) E

Q.100. An angle of 90° (Right angle) is inherently embedded in which geometric shape?

(A) Circle (B) Equilateral Triangle

(C) Right-angled Triangle (D) Regular Hexagon

SECTION D : GENERAL KNOWLEDGE (Q.101 TO 125)

Q.101. Which scientific device is used to measure the purity of milk?

(A) Barometer (B) Lactometer

(C) Thermometer (D) Hydrometer

Q.102. What is the official ratio of the width to the length of the Indian National Flag?

(A) 1 : 2 (B) 2 : 3

(C) 3 : 4 (D) 3 : 5

Q.103. Which classic folk dance originates from the state of Assam?

(A) Garba (B) Bihu

(C) Lavani (D) Ghoomar

Q.104. What is the highest gallantry award in India given for military bravery during wartime?

(A) Bharat Ratna
(B) Param Vir Chakra
(C) Maha Vir Chakra
(D) Shaurya Chakra

Q.105. Which organ in the human body produces bile juice that aids in fat digestion?

(A) Stomach
(B) Liver
(C) Pancreas
(D) Small Intestine

Q.106. Deficiency of Vitamin C in the human diet causes which disease?

(A) Beriberi
(B) Scurvy
(C) Rickets
(D) Goitre

Q.107. What scientific technique do bats use to navigate and catch prey in pitch darkness?

(A) Photosynthesis
(B) Echolocation
(C) Camouflage
(D) Hibernation

Q.108. Seeds of dandelion plants are dispersed primarily by which agent?

(A) Water
(B) Wind
(C) Animals
(D) Explosive mechanism

Q.109. Which traditional rainwater harvesting stepwell structure is widely found in Rajasthan?

(A) Johad
(B) Bawri (Baoli)
(C) Khadin
(D) Taanka

Q.110. Water boils at 100°C at sea level. What is this boiling point on the Kelvin scale?

(A) 273 K
(B) 373 K
(C) 100 K
(D) 300 K

Q.111. Which planet in our Solar System is commonly known as the 'Red Planet'?

(A) Venus
(B) Mars
(C) Jupiter
(D) Saturn

Q.112. The famous Sanchi Stupa, a World Heritage site, is situated in which Indian state?

(A) Uttar Pradesh
(B) Madhya Pradesh
(C) Bihar
(D) Rajasthan

Q.113. Who authored the famous Indian national song 'Vande Mataram'?

(A) Rabindranath Tagore
(B) Bankim Chandra Chattopadhyay
(C) Sarojini Naidu
(D) Sri Aurobindo

Q.114. What is the equivalent rank in the Indian Navy for a 'Colonel' in the Indian Army?

(A) Group Captain
(B) Captain
(C) Commodore
(D) Commander

Q.115. Which of the following is a non-renewable source of energy?

(A) Solar energy
(B) Coal
(C) Wind energy
(D) Hydroelectric power

Q.116. The World Health Organization (WHO) has its official headquarters in:

(A) New York
(B) Geneva
(C) Paris
(D) London

Q.117. What is the young one of a Kangaroo called?

(A) Calf
(B) Joey
(C) Cub
(D) Foal

Q.118. Which atmospheric process is primarily responsible for the formation of clouds?

(A) Evaporation
(B) Condensation
(C) Sublimation
(D) Transpiration

Q.119. The Durand Cup is one of Asia's oldest tournament associated with which sport?

(A) Cricket
(B) Football
(C) Hockey
(D) Badminton

Q.120. Which gas do green plants absorb from atmosphere during photosynthesis?

(A) Oxygen
(B) Carbon dioxide
(C) Nitrogen
(D) Hydrogen

Q.121. Who was the first Indian woman to win an individual medal at the Olympic Games?

(A) P.T. Usha
(B) Karnam Malleswari
(C) Mary Kom
(D) Saina Nehwal

Q.122. Which atmospheric layer contains the Ozone Layer that absorbs harmful UV rays?

(A) Troposphere
(B) Stratosphere
(C) Mesosphere
(D) Thermosphere

Q.123. Which type of soil is most suitable for growing Cotton crops in India?

(A) Alluvial soil
(B) Black soil (Regur)
(C) Red soil
(D) Desert soil

Q.124. The Kaziranga National Park in Assam is world famous for protecting which animal?

(A) Bengal Tiger
(B) Great One-horned Rhinoceros
(C) Asian Elephant
(D) Snow Leopard

Q.125. Which eminent Indian scientist received the Nobel Prize in Physics in 1930 for light scattering?

(A) Homi J. Bhabha
(B) C.V. Raman
(C) Jagadish Chandra Bose
(D) Vikram Sarabhai

PART B — ANSWER KEY (MOCK PAPER 01 : Q.1 TO 125)

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Q. No.	Opt	Answer	Q. No.	Opt	Answer	Q. No.	Opt	Answer	Q. No.	Opt	Answer
1	(B)	swarm	33	(A)	5/6, 4/5, 7/10, ..	65	(B)	5	97	(B)	Cube
2	(B)	since	34	(A)	13/400	66	(A)	6.83	98	(B)	43
3	(A)	a	35	(A)	18 days	67	(B)	6.25%	99	(C)	D
4	(B)	who	36	(B)	■ 1,200	68	(C)	Diameter	100	(C)	Right-angled Tri..
5	(B)	more costly	37	(C)	20%	69	(C)	Statement I is t..	101	(B)	Lactometer
6	(B)	are playing	38	(B)	■ 3,000	70	(C)	72	102	(B)	2 : 3
7	(B)	Accommodation	39	(D)	56 kg	71	(A)	■ 480 Gain	103	(B)	Bihu
8	(B)	Polite	40	(B)	520	72	(B)	5/8	104	(B)	Param Vir Chakra
9	(C)	Modern	41	(B)	■ 1,800	73	(B)	70 m	105	(B)	Liver
10	(B)	To study or work..	42	(B)	10 years	74	(B)	315 km	106	(B)	Scurvy
11	(C)	are you	43	(A)	57°	75	(C)	7	107	(B)	Echolocation
12	(A)	Spinster	44	(A)	80°	76	(C)	159	108	(B)	Wind
13	(B)	Hypotheses	45	(B)	60°	77	(C)	R	109	(B)	Bawri (Baoli)
14	(B)	kicked (Transiti..	46	(B)	74 cm	78	(B)	Mill	110	(B)	373 K
15	(C)	Adverb of Manner	47	(B)	208 m²	79	(B)	225	111	(B)	Mars
16	(B)	or	48	(B)	■ 1,320	80	(D)	Brass	112	(B)	Madhya Pradesh
17	(B)	SPQR	49	(B)	592 cm²	81	(D)	216	113	(B)	Bankim Chandra C..
18	(B)	Astronomer	50	(B)	192	82	(A)	JQTUG	114	(B)	Captain
19	(C)	Alas!	51	(B)	20 m/s	83	(B)	Mother	115	(B)	Coal
20	(B)	effect	52	(B)	60 km/hr	84	(B)	5 km	116	(B)	Geneva
21	(B)	Howrah, near Kol..	53	(C)	9	85	(B)	29th	117	(B)	Joey
22	(B)	They act like su..	54	(C)	40	86	(A)	5, 1, 4, 3, 2	118	(B)	Condensation
23	(B)	It was infected ..	55	(C)	30	87	(A)	2, 4, 3, 1	119	(B)	Football
24	(C)	Two severe cyclo..	56	(B)	113°F	88	(B)	22	120	(B)	Carbon dioxide
25	(C)	Thrive	57	(B)	+40	89	(D)	STREAM	121	(B)	Karnam Malleswari
26	(A)	86,931	58	(B)	+9	90	(B)	36	122	(B)	Stratosphere
27	(A)	80,008	59	(B)	7	91	(A)	K I N I A S (fli..	123	(B)	Black soil (Regur)
28	(B)	178	60	(A)	38,412	92	(A)	Two separate cir..	124	(B)	Great One-horned..
29	(B)	42	61	(B)	52 cm	93	(C)	8	125	(B)	C.V. Raman
30	(B)	72 seconds	62	(B)	26	94	(B)	Wednesday			
31	(A)	31	63	(B)	49,000	95	(B)	Q			
32	(A)	2 4/9	64	(C)	Reflex angle	96	(A)	All cats have lu..			

PART C — DETAILED STEP-BY-STEP SOLUTIONS (MOCK PAPER 01)

VIVEKANANDHA SAINIK COACHING CENTER

SOLUTIONS : SECTION A : ENGLISH LANGUAGE

Q.1. Correct Option: (B)
Explanation: A group of bees is called a 'swarm'. Therefore, option (B) is the correct collective noun.

Q.2. Correct Option: (B)
Explanation: 'Since' is used to denote a specific point of time in the past up to the present. Option (B) is correct.

Q.3. Correct Option: (A)
Explanation: 'Useful' starts with a consonant sound (/ju■stf■l/), so the indefinite article 'a' is used. Option (A) is correct.

Q.4. Correct Option: (B)
Explanation: 'Who' is used as a relative pronoun for persons in the nominative case. Option (B) is correct.

Q.5. Correct Option: (B)
Explanation: Comparative degree 'more costly' is used when comparing two items (platinum and gold). Option (B) is correct.

Q.6. Correct Option: (B)
Explanation: Action happening right now at the time of speaking requires Present Continuous tense ('are playing'). Option (B) is correct.

Q.7. Correct Option: (B)
Explanation: The correct spelling is 'Accommodation' (with double 'c' and double 'm'). Option (B) is correct.

Q.8. Correct Option: (B)
Explanation: 'Courteous' means showing polite behavior and respect towards others. Synonym is 'Polite' (B).

Q.9. Correct Option: (C)
Explanation: 'Ancient' means belonging to the distant past; its antonym is 'Modern' (C).

Q.10. Correct Option: (B)
Explanation: 'Burn the midnight oil' means to stay up late reading, studying, or working. Option (B) is correct.

Q.11. Correct Option: (C)
Explanation: A negative statement takes a positive question tag ('are you?'). Option (C) is correct.

Q.12. Correct Option: (A)
Explanation: The feminine counterpart of an unmarried man ('Bachelor') is 'Spinster'. Option (A) is correct.

Q.13. Correct Option: (B)
Explanation: Nouns ending in '-is' form their plural by changing '-is' to '-es' ('Hypotheses'). Option (B) is correct.

Q.14. Correct Option: (B)
Explanation: 'Kicked' requires a direct object ('football') to complete its sense, making it a Transitive Verb (B).

Q.15. Correct Option: (C)
Explanation: 'Bravely' answers the question 'How did the soldier fight?', so it is an Adverb of Manner (C).

Q.16. Correct Option: (B)
Explanation: 'Or' is used to express an alternative or consequence of not doing an action. Option (B) is correct.

Q.17. Correct Option: (B)
Explanation: Correct sentence order is S (He) + P (always) + Q (speaks) + R (the truth) = SPQR (Option B).

Q.18. Correct Option: (B)
Explanation: A scientist who studies celestial objects is an 'Astronomer'. Option (B) is correct.

Q.19. Correct Option: (C)
Explanation: 'Alas!' is an interjection used to express sorrow, grief, or regret. Option (C) is correct.

Q.20. Correct Option: (B)
Explanation: 'Effect' is a noun meaning a change or result brought about by a cause. Option (B) is correct.

Q.21. Correct Option: (B)
Explanation: The passage states that the tree is located in Howrah, near Kolkata. Option (B) is correct.

Q.22. Correct Option: (B)
Explanation: The passage mentions prop roots 'acting like supporting pillars'. Option (B) is correct.

Q.23. Correct Option: (B)
Explanation: The passage states the main trunk was infected by a deadly fungus after cyclones. Option (B) is correct.

Q.24. Correct Option: (C)
Explanation: The passage attributes the damage to two severe cyclones in 1864 and 1867. Option (C) is correct.

Q.25. Correct Option: (C)
Explanation: 'Thrive' means to grow or develop well and vigorously. Option (C) is correct.

SOLUTIONS : SECTION B : MATHEMATICS

Q.26. Correct Option: (A)
Explanation: Greatest number = 97,410. Smallest number = 10,479. Difference = 97,410 – 10,479 = 86,931 (A).

Q.27. Correct Option: (A)
Explanation: Place value of 8 = 80,000. Face value of 8 = 8. Sum = 80,000 + 8 = 80,008 (A).

Q.28. Correct Option: (B)
Explanation: C = 100, L = 50, XX = 20, V = 5, III = 3. Total = 100 + 50 + 20 + 5 + 3 = 178 (B).

Q.29. Correct Option: (B)
Explanation: 84 = $2^2 \times 3 \times 7$, 126 = $2 \times 3^2 \times 7$, 210 = $2 \times 3 \times 5 \times 7$. HCF = 2 × 3 × 7 = 42 (B).

Q.30. Correct Option: (B)
Explanation: Required time = LCM(6, 8, 12, 18). Prime factorization gives $2^3 \times 3^2$ = 72 seconds (B).

Q.31. Correct Option: (A)
Explanation: 12 ÷ 4 × 2 = 3 × 2 = 6. Then 30 – 6 = 24. Then 20 + 24 = 44. Finally 75 – 44 = 31 (A).

Q.32. Correct Option: (A)
Explanation: Numerator = 6/10 + 5/10 = 11/10. Denominator = 14/20 – 5/20 = 9/20. (11/10) ÷ (9/20) = (11/10) × (20/9) = 22/9 = 2 4/9 (A).

Q.33. Correct Option: (A)
Explanation: Decimals: 5/6 ≈ 0.833, 4/5 = 0.800, 7/10 = 0.700, 2/3 ≈ 0.667. Descending order: 5/6 > 4/5 > 7/10 > 2/3 (A).

Q.34. Correct Option: (A)
Explanation: 0.0325 = 325 / 10,000. Dividing numerator and denominator by 25 gives 13 / 400 (A).

Q.35. Correct Option: (A)
Explanation: Inverse proportion: Workers × Days = Constant. 15 × 24 = 360 worker-days. Days = 360 / 20 = 18 days (A).

Q.36. Correct Option: (B)
Explanation: Total parts = 3 + 4 + 5 = 12. B's share = (4 / 12) × ■ 3,600 = ■ 1,200 (B).

Q.37. Correct Option: (C)
Explanation: Gain = SP – CP = 540 – 450 = ■ 90. Gain % = (90 / 450) × 100 = 20% (C).

Q.38. Correct Option: (B)
Explanation: SP = CP × (100 – Loss%)/100 => 2850 = CP × 0.95 => CP = 2850 / 0.95 = ■ 3,000 (B).

Q.39. Correct Option: (D)
Explanation: Total weight of 6 boys = 6 × 42 = 252 kg. Total weight of 7 boys = 7 × 44 = 308 kg. New boy = 308 – 252 = 56 kg (D).

Q.40. Correct Option: (B)
Explanation: Percentage of boys = 100% – 35% = 65%. Number of boys = 65% of 800 = 0.65 × 800 = 520 (B).

Q.41. Correct Option: (B)
Explanation: SI = (P × R × T) / 100 = (8000 × 7.5 × 3) / 100 = 80 × 22.5 = ■ 1,800 (B).

Q.42. Correct Option: (B)
Explanation: To double, Interest = Principal = ■ 5,000. Time = (SI × 100) / (P × R) = (5000 × 100) / (5000 × 10) = 10 years (B).

Q.43. Correct Option: (A)
Explanation: Let angle be x. Complement = 90° – x. x = (90° – x) + 24° => 2x = 114° => x = 57° (A).

Q.44. Correct Option: (A)
Explanation: Sum = 180°. Total parts = 4 + 5 = 9. Smaller angle = (4/9) × 180° = 80° (A).

Q.45. Correct Option: (B)
Explanation: Sum of angles in a triangle = 180°. ∠C = 180° – (55° + 65°) = 180° – 120° = 60° (B).

Q.46. Correct Option: (B)
Explanation: Breadth = Area / Length = 300 / 25 = 12 cm. Perimeter = 2 × (25 + 12) = 2 × 37 = 74 cm (B).

Q.47. Correct Option: (B)
Explanation: Inner side = 24 m (Area = 576 m^2). Outer side = 24 + 2 + 2 = 28 m (Area = 784 m^2). Path area = 784 – 576 = 208 m^2 (B).

Q.48. Correct Option: (B)
Explanation: Circumference = 2 × (22/7) × 14 = 88 m. Total cost = 88 × ■ 15 = ■ 1,320 (B).

Q.49. Correct Option: (B)
Explanation: TSA = 2(lb + bh + hl) = 2(12×10 + 10×8 + 8×12) = 2(120 + 80 + 96) = 2(296) = 592 cm² (B).

Q.50. Correct Option: (B)
Explanation: Volume of cuboid = 32×24×16 = 12,288 cm³. Volume of cube = 4³ = 64 cm³. Number of cubes = 12288 / 64 = 192 (B).

Q.51. Correct Option: (B)
Explanation: Speed in m/s = 72 × (5 / 18) = 4 × 5 = 20 m/s (B).

Q.52. Correct Option: (B)
Explanation: Speed = Distance / Time = 150 / 9 = 50/3 m/s. In km/hr = (50/3) × (18/5) = 60 km/hr (B).

Q.53. Correct Option: (C)
Explanation: 5x – 3x = 11 + 7 => 2x = 18 => x = 9 (C).

Q.54. Correct Option: (C)
Explanation: 2(3)² – 3(3)(–2) + (–2)² = 2(9) – (–18) + 4 = 18 + 18 + 4 = 40 (C).

Q.55. Correct Option: (C)
Explanation: Let numbers be n–1, n, n+1. Sum = 3n = 87 => n = 29. Numbers are 28, 29, 30. Greatest = 30 (C).

Q.56. Correct Option: (B)
Explanation: F = (9/5 × C) + 32 = (9/5 × 45) + 32 = 81 + 32 = 113°F (B).

Q.57. Correct Option: (B)
Explanation: Negative sum = –38 – 27 = –65. Positive sum = 45 + 60 = +105. Total = 105 – 65 = +40 (B).

Q.58. Correct Option: (B)
Explanation: (–15) – (–24) = –15 + 24 = +9 (B).

Q.59. Correct Option: (B)
Explanation: Prime numbers between 40 and 70 are: 41, 43, 47, 53, 59, 61, 67 (Total 7 numbers) (B).

Q.60. Correct Option: (A)
Explanation: In 38,412: Sum of odd digits = 3+4+2=9. Sum of even digits = 8+1=9. Difference = 0 (Divisible by 11) (A).

Q.61. Correct Option: (B)
Explanation: Perimeter of regular octagon = 8 × side = 8 × 6.5 cm = 52 cm (B).

Q.62. Correct Option: (B)
Explanation: 6.5 L = 6,500 mL. Number of glasses = 6,500 / 250 = 26 (B).

Q.63. Correct Option: (B)
Explanation: Digit at hundreds place is 7 (≥ 5), so round up to 49,000 (B).

Q.64. Correct Option: (C)
Explanation: An angle between 180° and 360° is called a Reflex Angle (C).

Q.65. Correct Option: (B)
Explanation: A regular polygon with n sides has n lines of symmetry. Regular pentagon has 5 lines of symmetry (B).

Q.66. Correct Option: (A)
Explanation: First 6 primes = 2, 3, 5, 7, 11, 13. Sum = 41. Mean = 41 / 6 ≈ 6.83 (A).

Q.67. Correct Option: (B)
Explanation: 4 kg = 4,000 g. Percentage = (250 / 4000) × 100 = 25/4 = 6.25% (B).

Q.68. Correct Option: (C)
Explanation: The longest chord passing through the center of a circle is its Diameter (C).

Q.69. Correct Option: (C)
Explanation: Statement I is true (a square has 4 right angles). Statement II is false (rhombus diagonals bisect at 90° but are unequal unless square) (C).

Q.70. Correct Option: (C)
Explanation: Numbers are 5 × 9 = 45 and 8 × 9 = 72. Larger number is 72 (C).

Q.71. Correct Option: (A)
Explanation: CP = ■ 2,400. SP = 40 × 72 = ■ 2,880. Gain = 2,880 – 2,400 = ■ 480 Gain (A).

Q.72. Correct Option: (B)
Explanation: LCM of 8, 12, 6 = 24. (9 + 10 – 4) / 24 = 15 / 24 = 5 / 8 (B).

Q.73. Correct Option: (B)
Explanation: 2(7x + 4x) = 220 => 22x = 220 => x = 10. Length = 7 × 10 = 70 m (B).

Q.74. Correct Option: (B)
Explanation: Speed = 180 / 4 = 45 km/hr. Distance in 7 hours = 45 × 7 = 315 km (B).

Q.75. Correct Option: (C)
Explanation: Number of symbols = 105 / 15 = 7 symbols (C).

SOLUTIONS : SECTION C : INTELLIGENCE

Q.76. Correct Option: (C)
Explanation: Pattern: Previous term × 2 + 1. Next term = 79 × 2 + 1 = 158 + 1 = 159 (C).

Q.77. Correct Option: (C)
Explanation: Pattern: +3 position increment in alphabet (3, 6, 9, 12, 15, 18). 18th letter is R (C).

Q.78. Correct Option: (B)
Explanation: Newspaper is produced in a Press; similarly, Cloth is produced in a Mill (B).

Q.79. Correct Option: (B)
Explanation: Relationship is n : n^2. $12^2 = 144$, so $15^2 = 225$ (B).

Q.80. Correct Option: (D)
Explanation: Copper, Silver, and Gold are pure metallic elements; Brass is an alloy (mixture of copper and zinc) (D).

Q.81. Correct Option: (D)
Explanation: 121 (11^2), 169 (13^2), and 225 (15^2) are perfect squares; 216 is a perfect cube (6^3) (D).

Q.82. Correct Option: (A)
Explanation: Each letter is moved +2 forward: H+2=J, O+2=Q, R+2=T, S+2=U, E+2=G -> JQTUG (A).

Q.83. Correct Option: (B)
Explanation: 'Only daughter of my mother' is Neha herself. Therefore, Neha is his Mother (B).

Q.84. Correct Option: (B)
Explanation: Initial movement 10 km East and 10 km West cancel out. He is 5 km South from starting point (B).

Q.85. Correct Option: (B)
Explanation: Rank from bottom = Total – Top rank + 1 = 40 – 12 + 1 = 29th (B).

Q.86. Correct Option: (A)
Explanation: Dictionary order: Magnet (5), Mango (1), Margin (4), Market (3), Master (2) -> 5, 1, 4, 3, 2 (A).

Q.87. Correct Option: (A)
Explanation: Smallest to largest: Millimetre (2) < Centimetre (4) < Metre (3) < Kilometre (1) -> 2, 4, 3, 1 (A).

Q.88. Correct Option: (B)
Explanation: Substitute symbols: 36 ÷ 6 × 3 + 8 – 4 = 6 × 3 + 8 – 4 = 18 + 8 – 4 = 22 (B).

Q.89. Correct Option: (D)
Explanation: The word TRANSPARENT does not contain the letter 'M', so STREAM cannot be formed (D).

Q.90. Correct Option: (B)
Explanation: Pattern: First × Second + 1. Row 1: 3×5+1=16. Row 2: 4×6+1=25. Row 3: 5×7+1 = 36 (B).

Q.91. Correct Option: (A)
Explanation: Mirror image reverses letter order right-to-left and flips individual characters horizontally (A).

Q.92. Correct Option: (A)
Explanation: Potato and Cabbage are distinct items, both belonging entirely to the group Vegetables (A).

Q.93. Correct Option: (C)
Explanation: 4 small individual triangles + 4 larger triangles formed by diagonal pairs = 8 triangles (C).

Q.94. Correct Option: (B)
Explanation: Difference = 30 days. 30 ÷ 7 = 4 weeks + 2 odd days. Monday + 2 days = Wednesday (B).

Q.95. Correct Option: (B)
Explanation: 12th letter from left is L. 5th right from L is 12 + 5 = 17th letter, which is Q (B).

Q.96. Correct Option: (A)
Explanation: By categorical syllogism, if all A are B and all B are C, then all A are C (A).

Q.97. Correct Option: (B)
Explanation: A circle is 2D and sphere is its 3D counterpart. Square is 2D and Cube is its 3D counterpart (B).

Q.98. Correct Option: (B)
Explanation: E=5, X=24, A=1, M=13. Sum = 5 + 24 + 1 + 13 = 43 (B).

Q.99. Correct Option: (C)
Explanation: Order from left to right: E, B, C, A, D. D is sitting at the extreme right end (C).

Q.100. Correct Option: (C)
Explanation: A right-angled triangle contains one angle measuring exactly 90° (C).

SOLUTIONS : SECTION D : GENERAL KNOWLEDGE

Q.101. Correct Option: (B)
Explanation: A Lactometer measures the specific gravity of milk to test its purity (B).

Q.102. Correct Option: (B)
Explanation: The ratio of the width (height) to the length of the Indian Tricolour flag is 2 : 3 (B).

Q.103. Correct Option: (B)
Explanation: Bihu is a famous traditional folk dance and festival of Assam (B).

Q.104. Correct Option: (B)
Explanation: The Param Vir Chakra (PVC) is India's highest wartime military decor (B).

Q.105. Correct Option: (B)
Explanation: The Liver secretes bile juice, which is stored in the gallbladder and digests fats (B).

Q.106. Correct Option: (B)
Explanation: Vitamin C deficiency causes Scurvy, characterized by bleeding gums and slow wound healing (B).

Q.107. Correct Option: (B)
Explanation: Bats emit high-frequency ultrasonic sounds and listen to echoes to locate objects (Echolocation) (B).

Q.108. Correct Option: (B)
Explanation: Dandelion seeds have light hair-like structures (parachutes) dispersed by Wind (B).

Q.109. Correct Option: (B)
Explanation: A Bawri is a traditional stepwell built to harvest rainwater in arid regions (B).

Q.110. Correct Option: (B)
Explanation: Kelvin = Celsius + 273. 100°C + 273 = 373 K (B).

Q.111. Correct Option: (B)
Explanation: Mars is called the Red Planet due to iron oxide (rust) on its surface (B).

Q.112. Correct Option: (B)
Explanation: Sanchi Stupa is an ancient Buddhist monument in Raisen district, Madhya Pradesh (B).

Q.113. Correct Option: (B)
Explanation: 'Vande Mataram' was written in Sanskritized Bengali by Bankim Chandra Chattopadhyay in his novel Anandamath (B).

Q.114. Correct Option: (B)
Explanation: A Colonel in the Army is equivalent in rank to a Captain in the Navy and Group Captain in the Air Force (B).

Q.115. Correct Option: (B)
Explanation: Coal is a fossil fuel that depletes over time and cannot be renewed quickly (B).

Q.116. Correct Option: (B)
Explanation: WHO is a specialized agency of the United Nations headquartered in Geneva, Switzerland (B).

Q.117. Correct Option: (B)
Explanation: A young marsupial, including a kangaroo, is called a Joey (B).

Q.118. Correct Option: (B)
Explanation: Water vapour rises, cools, and condenses into water droplets to form clouds (Condensation) (B).

Q.119. Correct Option: (B)
Explanation: The Durand Cup is a prestigious historic football tournament in India (B).

Q.120. Correct Option: (B)
Explanation: Plants absorb Carbon dioxide (CO■) and release Oxygen (O■) during photosynthesis (B).

Q.121. Correct Option: (B)
Explanation: Karnam Malleswari won a Bronze medal in Weightlifting at Sydney 2000 Olympics (B).

Q.122. Correct Option: (B)
Explanation: The Ozone Layer is located in the Stratosphere, about 15-35 km above Earth's surface (B).

Q.123. Correct Option: (B)
Explanation: Black soil (Regur) has high clay content and moisture retention, ideal for Cotton (B).

Q.124. Correct Option: (B)
Explanation: Kaziranga holds two-thirds of the world's Great One-horned Rhinoceroses (B).

Q.125. Correct Option: (B)
Explanation: Sir C.V. Raman was awarded the Nobel Prize for discovering the 'Raman Effect' (B).